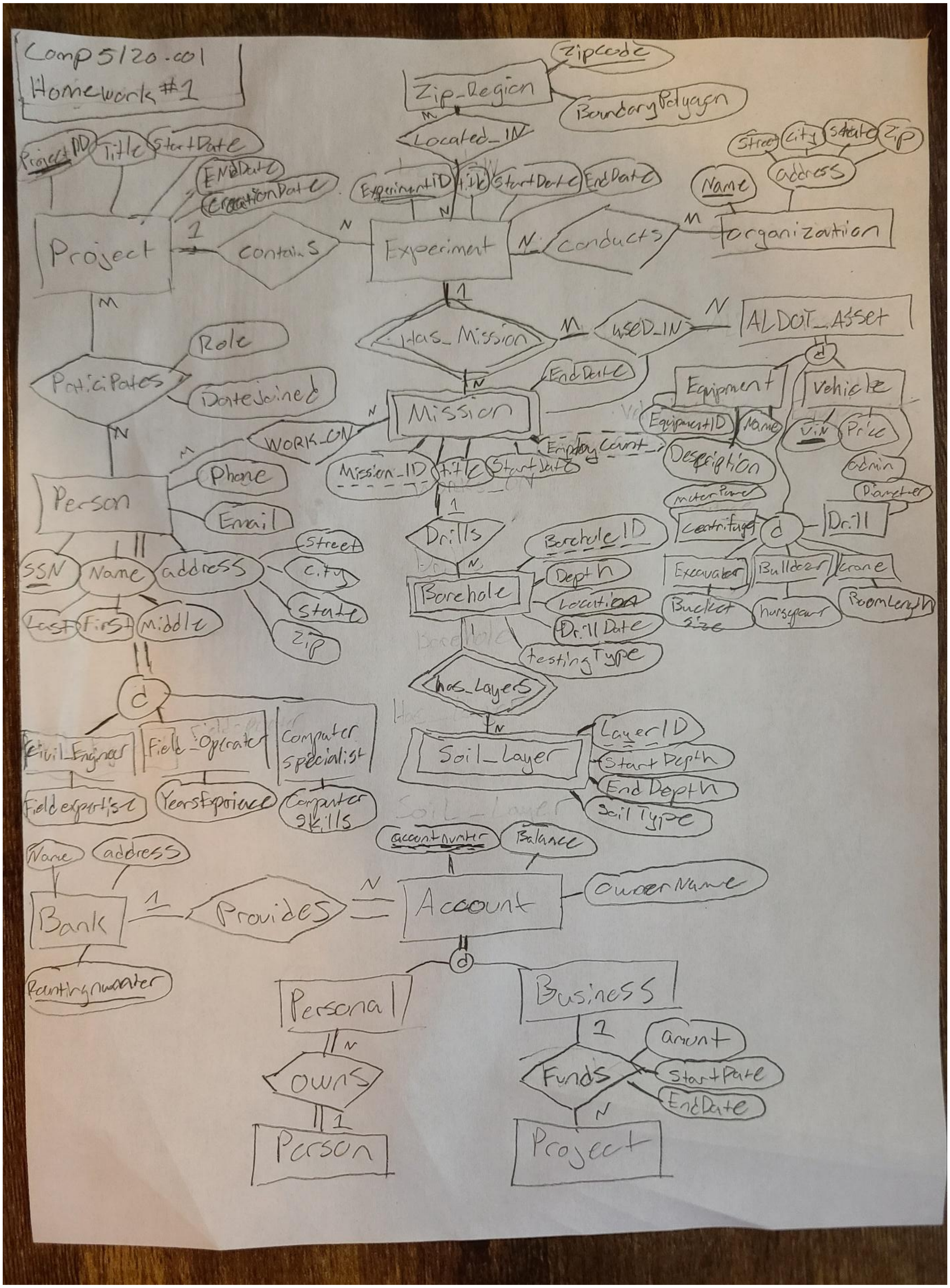
Comp 5120.001
Homework #1
Zip_Region
Zipcode
BoundaryPolygon
M
Located_IN
N
ProjectID
Title
StartDate
EndDate
CreationDate
ExperimentID
title
StartDate
EndDate
Street
city
state
Zip
address
Name
Project
1
contains
N
Experiment
N
conducts
M
organization
M
1
Has_Mission
M
used_IN
N
ALDOT_Asset
d
Role
Participates
DateJoined
N
EndDate
Mission
Equipment
Vehicle
N
WORK_ON
M
EquipmentID
Name
VIN
Price
Employee_Count
Description
admin
Mission_ID
title
StartDate
motorPower
Diameter
1
Drills
N
Centrifuge
d
Drill
Person
Phone
Email
BoreholeID
Borehole
Depth
Location
DrillDate
testingType
Excavator
Bulldozer
crane
Bucket size
horsepower
BoomLength
SSN
Name
address
Street
city
state
zip
Last
First
Middle
d
Has_Layers
N
Civil_Engineer
Field_Operator
Computer specialist
Fieldexpertise
YearsExperience
Computer skills
Soil_Layer
LayerID
Start Depth
End Depth
Soil Type
accountnumber
Balance
Name
address
Bank
1
Provides
N
Account
OwnerName
Routingnumber
d
Personal
Business
N
owns
1
Person
1
Funds
N
Project
amount
StartDate
EndDate

Modeling Assumptions

1. Each EXPERIMENT belongs to one PROJECT; a PROJECT may contain multiple EXPERIMENTs.
2. An EXPERIMENT may be associated with multiple ZIP_REGIONs, and a ZIP_REGION may contain multiple EXPERIMENTs.
3. Each EXPERIMENT is conducted by one ORGANIZATION; an ORGANIZATION may conduct multiple EXPERIMENTs.
4. Role and DateJoined are attributes of PERSON's participation in a PROJECT. Role is multivalued because a participant has at least one role.
5. CIVIL_ENGINEER, FIELD_OPERATOR, and COMPUTER_SPECIALIST are modeled as a total, disjoint specialization of PERSON.
6. MISSION is a weak entity identified by its owning EXPERIMENT together with partial key MissionID.
7. EmployeeCount is treated as a derived MISSION attribute because it can be obtained from WORKS_ON participation.
8. BOREHOLE is modeled as weak; BoreholeID is unique within the experiment. Each borehole is drilled for one MISSION.
9. SOIL_LAYER is modeled as weak under BOREHOLE, with LayerID as its partial key.
10. ALDOT_ASSET is introduced as a superclass so both VEHICLE and EQUIPMENT can participate in USED_IN with MISSION.
11. EXCAVATOR, BULLDOZER, CRANE, CENTRIFUGE, and DRILL are modeled as a partial, disjoint specialization of ALDOT_ASSET.
12. BANK PhoneNumber and COMPUTER_SPECIALIST ComputerSkills are multivalued attributes.
13. ACCOUNT is specialized totally and disjointly into PERSONAL_ACCOUNT and BUSINESS_ACCOUNT.
14. FUNDS is modeled as a relationship between BUSINESS_ACCOUNT and PROJECT with Amount, StartDate, and EndDate attributes.

Notation Used

Rectangle - Strong entity type
Double rectangle - Weak entity type
Diamond - Relationship type
Double diamond - Identifying relationship
Oval - Attribute
Underlined attribute - Key attribute
Dashed underlined attribute - Partial key of a weak entity
Double oval - Multivalued attribute
Dashed oval - Derived attribute
Double participation line - Total participation
1 / M / N - Relationship cardinality
d specialization circle - Disjoint specialization